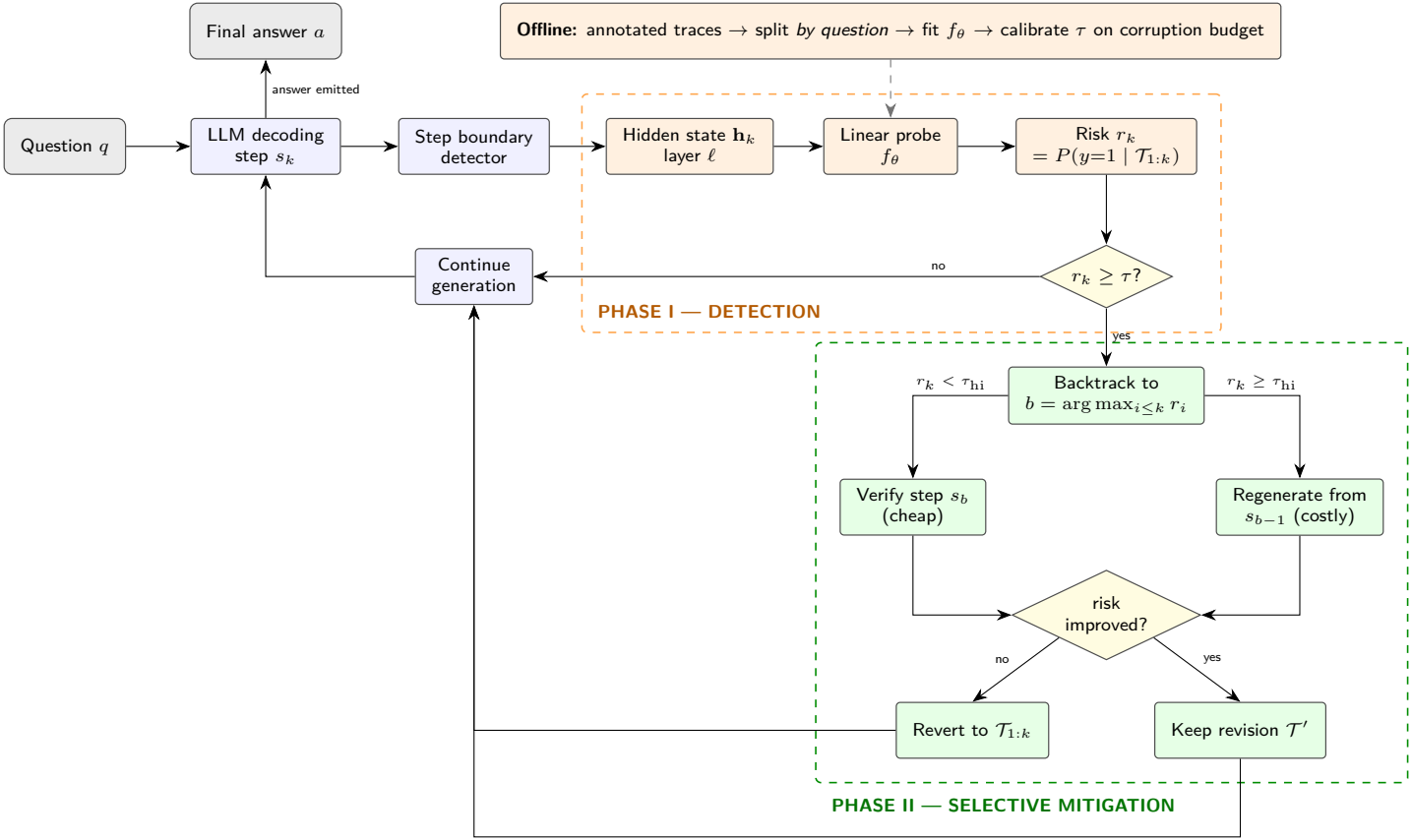
Final answer a
Offline: annotated traces → split by question → fit f_θ → calibrate τ on corruption budget
answer emitted
Question q
LLM decoding step s_k
Step boundary detector
Hidden state $\mathbf{h}_k$ layer ℓ
Linear probe f_θ
Risk r_k $= P(y{=}1 \mid \mathcal{T}_{1:k})$
$r_k \geq \tau$?
no
Continue generation
PHASE I — DETECTION
yes
$r_k < \tau_{\text{hi}}$
Backtrack to $b = \arg\max_{i \leq k} r_i$
$r_k \geq \tau_{\text{hi}}$
Verify step s_b (cheap)
Regenerate from s_{b-1} (costly)
risk improved?
no
yes
Revert to $\mathcal{T}_{1:k}$
Keep revision $\mathcal{T}'$
PHASE II — SELECTIVE MITIGATION